COUNTEREXAMPLES, SPECTRAL OBSTRUCTIONS, AND DELETION STABILITY FOR WOW-284

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Abstract. WOW-284 asserts that the minimum dual degree of every connected graph of order at least three and girth at least five does not exceed the negative of its least distance eigenvalue. We refute it with exact counterexamples of orders $38, 39, 40, 42$, and 50, and develop a structural theory of the failure. For a connected k-regular graph of girth at least five and diameter three, we prove

$$\delta^*(G) + \lambda_{\min}(D(G)) = 2k - 2 - \max_{\theta \neq k}(\theta + 1)^2.$$

Consequently every regular strict counterexample has degree at least six and diameter at most four, while diameter four forces degree at least ten. We solve the associated one-variable nonbacktracking linear program exactly, including optimizer rigidity; an edge-local cycle-divisibility sieve excludes degree-six order 51, so degree-six counterexamples have order at most 50. We also determine the distance spectra of one- and two-vertex punctures of Moore graphs and establish a uniform deletion-stability bound: every deletion of at most five vertices from the Hoffman–Singleton graph remains a strict counterexample, whereas an explicit six-vertex deletion does not. All theorem-level computations use exact arithmetic. Lean 4.31 kernel-checks the full 50-vertex proof, finite spectral certificates at orders $38, 39, 40, 42$, and the analytic LP optimum and rigidity for every integer $k \geq 4$.

1. Introduction

Let G be a finite simple connected graph. Write $d(v)$ for the degree of v, $N(v)$ for its open neighbourhood, and, for $S \subseteq V(G)$, let $G - S$ denote the subgraph induced by $V(G) \setminus S$. Define

$$d^*(v) = \frac{1}{d(v)} \sum_{u \in N(v)} d(u), \qquad \delta^*(G) = \min_{v \in V(G)} d^*(v).$$

The distance matrix is $D(G) = (d_G(u, v))_{u,v \in V(G)}$. Its eigenvalues are ordered as

$$\partial_1(G) \geq \cdots \geq \partial_n(G), \qquad \partial_n(G) = \lambda_{\min}(D(G)).$$

Aouchiche and Hansen record the following Graffiti conjecture as Conjecture 7.16 and attribute it to Fajtlowicz's 1998 *Written on the Wall* report [1] (see also [2, Conjecture 7.16]).

Conjecture (WOW-284). *If G has order at least three and girth at least five, then*

$$\delta^*(G) \leq -\lambda_{\min}(D(G)).$$

For graphs in the domain of the conjecture, put

$$\Phi(G) = \delta^*(G) + \lambda_{\min}(D(G)).$$

Thus G is a strict counterexample precisely when $\Phi(G) > 0$.

The initial disproof is short. A degree-k Moore graph of diameter two has

$$A^2 = (k - 1)I - A + J, \qquad D = 2J - 2I - A,$$

and hence

$$\delta^*(G) = k, \qquad \lambda_{\min}(D) = -\frac{3 + \sqrt{4k - 3}}{2}.$$

The conjecture holds on these graphs exactly for $k \leq 3$, with equality at $k = 3$, and fails for every realizable $k > 3$. The degree-seven Hoffman–Singleton graph therefore gives a gap of three.

The purpose of this paper is not merely to list descendants of this graph. It addresses three structural questions.

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(1) Which spectral mechanism governs regular counterexamples?
(2) How restrictive are the degree, diameter, and order conditions?
(3) How stable is the counterexample property under deletion?

Our main conclusions are as follows.

- In diameter three, WOW-284 is equivalent to a two-sided adjacency spectrum window centred at -1; see Theorem 3.1.
- Every regular strict counterexample has degree at least six and diameter at most four. A diameter-four example, if one exists, has degree at least ten; see Theorems 4.2, 4.3, and 4.4.
- The standard one-variable nonbacktracking linear-programming hierarchy has exact ceiling
$$B_k = \frac{(k+2)(k^2+3)}{6},$$
with a unique optimizer up to positive scaling; see Theorem 5.3.
- The LP ceiling gives $n \leq 51$ at degree six. An edge-local positive-semidefinite certificate yields a general cycle-divisibility sieve and rules out $n = 51$ by the congruence
$$5N_5 = 153 \cdot 11 = 1683,$$
so $n \leq 50$; see Theorem 6.3.
- One-vertex, adjacent-pair, and nonadjacent-pair deletions of Moore graphs admit exact invariant-subspace decompositions for their recomputed distance matrices; see Section 7.
- Every deletion of at most five vertices from the Hoffman–Singleton graph remains a strict counterexample, and this universal radius is sharp; see Theorem 8.3.

The proofs form two complementary hierarchies. In the obstruction direction, the distance-polynomial identity converts WOW-284 into a shifted adjacency window; scalar trace moments give the one-point LP ceiling; 2×2 principal minors localize that certificate to edge–cycle counts; and 3×3 minors impose the two-path constraints at order fifty. In the stability direction, deleting vertices from a Moore graph produces an explicit Gram-type correction to the distance matrix. Invariant-subspace decompositions then give exact puncture spectra, while orbitwise positive-definiteness certificates determine the sharp Hoffman–Singleton deletion radius.

The distance-polynomial viewpoint is established in the theory of minimal cages, distance-polynomial graphs, and spectral excess [3, 4]. Nonbacktracking linear-programming bounds are due to Nozaki and related spectral-Moore work [5, 6]. Our contribution is the specialization to the two-sided WOW window, the exact method ceiling, the edge-local cycle certificate, and the deletion theory developed below. Completed in its original counterexample form on 19 July 2026, this work is, to the author's knowledge, the first proof that WOW-284 is false and the first Lean formalization of such a refutation. We do not claim that order 38 is minimum or that the constructions classify all counterexamples. The literature boundaries are returned to in Section 11.

The analytic arguments are proved in the text. Precisely specified finite classifications and matrix certificates are treated as computer-assisted proof components. Their exact reproducibility materials are archived with the accompanying release.

2. Local growth, Moore graphs, and explicit counterexamples

2.1. **Dual degree as radius-two growth.** For $v \in V(G)$, write $\Gamma_i(v)$ for the distance-i sphere and $B_2(v) = \Gamma_0(v) \cup \Gamma_1(v) \cup \Gamma_2(v)$.

Proposition 2.1 (Backelin's second-degree identity). *If G contains no triangle and no 4-cycle, then*
$$|B_2(v)| = 1 + \sum_{u \in N(v)} d(u), \qquad d^*(v) = \frac{|B_2(v)| - 1}{d(v)}.$$

Proof. For distinct neighbours u, w of v, the sets $N(u) \setminus \{v\}$ and $N(w) \setminus \{v\}$ are disjoint; an intersection would form a 4-cycle, and a member in $N(v)$ would form a triangle. These sets partition $\Gamma_2(v)$, so
$$|\Gamma_2(v)| = \sum_{u \in N(v)} (d(u) - 1).$$

Adding the centre and first sphere proves the formula. This is the normalized form of Backelin's second-degree lemma [7, Lemma 2.1]. □

2.2. **The Moore threshold.**

Theorem 2.2. *Let M be a degree-k Moore graph of diameter two, $k \geq 2$. Then*

$$|V(M)| = k^2 + 1, \qquad g(M) = 5, \qquad \delta^*(M) = k,$$

$$\lambda_{\min}(D(M)) = -\frac{3 + \sqrt{4k-3}}{2},$$

and

$$\Phi(M) = k - \frac{3 + \sqrt{4k-3}}{2}.$$

Thus M satisfies WOW-284 exactly for $k \leq 3$, with equality exactly at $k = 3$.

Proof. The Moore bound is attained, so adjacent vertices have no common neighbour and nonadjacent vertices have exactly one. Given an edge uv, choose $x \in N(u) \setminus \{v\}$ and $y \in N(v) \setminus \{u\}$. The vertices x, y are nonadjacent, since an edge would create a four-cycle, and their unique common neighbour completes a five-cycle through uv. Therefore

$$A^2 = (k-1)I - A + J.$$

On $\mathbf{1}^\perp$, the nonprincipal adjacency eigenvalues are the roots

$$r, s = \frac{-1 \pm \sqrt{4k-3}}{2}.$$

Every nonedge has distance two, hence $D = 2J - 2I - A$. The least distance eigenvalue is $-2 - r = -(3 + \sqrt{4k-3})/2$. Regularity gives $\delta^* = k$, and

$$(2k-3)^2 - (4k-3) = 4(k-1)(k-3)$$

gives the threshold. The exact scalar and finite checks are independently repeated by `scripts/verify_regular_score_calculus.py`. □

2.3. **A coordinate Hoffman–Singleton certificate.**

All subscripts below lie in $\mathbb{F}_5 = \mathbb{Z}/5\mathbb{Z}$. Let

$$V(M) = \{P_{i,j} : i, j \in \mathbb{F}_5\} \,\dot\cup\, \{Q_{k,\ell} : k, \ell \in \mathbb{F}_5\},$$

with edges

$$\begin{aligned} P_{i,j} &\sim P_{i,j\pm1}, \\ Q_{k,\ell} &\sim Q_{k,\ell\pm2}, \\ P_{i,j} &\sim Q_{k,ik+j}. \end{aligned}$$

This is Hafner's affine-coordinate form of the Hoffman–Singleton graph after a minor reindexing [8].

Proposition 2.3. *The coordinate construction is a simple connected 7-regular graph on 50 vertices. Adjacent pairs have no common neighbour and nonadjacent pairs have exactly one. Consequently it has girth five, diameter two, and*

$$\operatorname{Spec} D(M) = \{91^{(1)}, 1^{(21)}, (-4)^{(28)}\}.$$

In particular, $\delta^*(M) = 7$ and $\Phi(M) = 3$.

Proof. The neighbourhoods are

$$\begin{aligned} N(P_{i,j}) &= \{P_{i,j-1}, P_{i,j+1}\} \cup \{Q_{k,ik+j} : k \in \mathbb{F}_5\}, \\ N(Q_{k,\ell}) &= \{Q_{k,\ell-2}, Q_{k,\ell+2}\} \cup \{P_{i,\ell-ik} : i \in \mathbb{F}_5\}. \end{aligned}$$

They have seven distinct entries. For two vertices of the same type, common neighbours are determined by one nonzero linear equation over $\mathbb{F}_5$; for a cross pair $P_{i,j}, Q_{k,\ell}$, put $r = \ell - (ik + j)$. The pair is adjacent for $r = 0$, has one common P-neighbour for $r \in \{\pm1\}$, and one common Q-neighbour for $r \in \{\pm2\}$. The five residues are exhausted. The claimed geometry and spectrum now follow from Theorem 2.2. The exhaustive pair certificate, integer BFS distances, characteristic polynomial, and exact positive-definiteness check are in `scripts/verify_exact.py`. □

2.4. **Smaller exact counterexamples.** Let

$$\mathcal{P} = \{P_{0,j}, Q_{0,j} : j \in \mathbb{F}_5\}.$$

The induced graph $M[\mathcal{P}]$ is a Petersen graph. Put

$$R = M - \mathcal{P}, \quad H_{39} = R - P_{1,0}, \quad H_{38} = R - \{P_{1,0}, P_{1,1}\},$$

and let X_{42} be the second subconstituent of $P_{0,0}$, namely the graph induced by the vertices at distance two from it.

Theorem 2.4. *The following are strict counterexamples.*

G	$\lvert V(G)\rvert$	$\delta^*(G)$	$\lambda_{\min}(D(G))$
H_{38}	38	$17/3$	$-3-\sqrt{7}$
H_{39}	39	$35/6$	$> -35/6$
R	40	6	-5
X_{42}	42	6	-5
M	50	7	-4

The entry for H_{39} records an exact strict lower bound obtained from positive definiteness of $6D + 35I$; it is not a decimal approximation to the least eigenvalue. Moreover, all 40 labelled singleton deletions of R, and all 120 labelled deletions of the endpoints of an edge of R, are strict counterexamples with the same respective characteristic polynomials.

Proof. The graph R is the classical $(6,5)$-cage of O'Keefe and Wong [9, 10]; its realization as a Petersen deletion of the Hoffman–Singleton graph also appears in [11]. The Moore block identity gives

$$\operatorname{Spec} A(R) = \{6^{(1)}, 2^{(18)}, 1^{(4)}, (-2)^{(5)}, (-3)^{(12)}\},$$

and Theorem 3.1 below maps this to

$$\operatorname{Spec} D(R) = \{75^{(1)}, 3^{(5)}, 0^{(16)}, (-5)^{(18)}\}.$$

The second-subconstituent calculation gives

$$\operatorname{Spec} D(X_{42}) = \{81^{(1)}, 4^{(6)}, 0^{(14)}, (-5)^{(21)}\}.$$

The classical second-subconstituent identification and adjacency spectrum are recorded in [12]. For H_{38}, a direct degree count gives $\delta^* = 17/3$, while the factor $x^2 + 6x + 2$, together with an exact Sturm isolation, gives the least root $-3-\sqrt{7}$. For H_{39}, the exact matrix $6D+35I$ is positive definite. All graph, girth, distance, dual-degree, Sturm, and rational LDL^{T} certificates are checked by `scripts/verify_extended.py`; the labelled deletion families are exhausted by `scripts/verify_descendant_families.py`. No transitivity or numerical root ordering is used. □

Theorem 2.5 (Moore second subconstituents)**.** *Let M be a degree-K Moore graph of diameter two, $K \geq 3$, and let X be the graph induced by $\Gamma_2(v)$ for a fixed vertex v. Then X has order $K(K-1)$, degree $K-1$, girth at least five, diameter three, and*

$$\lambda_{\min}(D(X)) = -\frac{5+\sqrt{4K-3}}{2}.$$

Consequently

$$\Phi(X) = K - 1 - \frac{5+\sqrt{4K-3}}{2},$$

which is positive exactly for integers $K \geq 6$.

Proof. Relative to $\{v\} \sqcup N(v) \sqcup \Gamma_2(v)$, write the adjacency matrix with incidence block C and induced block B. The Moore identity gives

$$CC^{\mathsf{T}} = (K-1)I, \qquad CB = J - C,$$

and on $\ker C$,

$$B^2 + B - (K-1)I = 0.$$

Since $CC^{\mathsf{T}} = (K-1)I$, the map C^{T} is injective, its image is orthogonal to $\ker C$, and

$$\mathbb{R}^{\Gamma_2(v)} = \langle \mathbf{1} \rangle \perp C^{\mathsf{T}}(\mathbf{1}^{\perp}) \perp \ker C.$$

Thus B has principal eigenvalue $K-1$, eigenvalue -1 on $C^{\mathsf{T}}(\mathbf{1}^{\perp})$, and the two Moore roots $(-1\pm\sqrt{4K-3})/2$ on $\ker C$. It remains to identify the diameter of X. If the unique common neighbour in M of two nonadjacent vertices $x, y \in X$ lies in X, their distance in X is two. Otherwise it is their common parent in $N(v)$. Choose $b \in N_X(x)$. Then $b \nsim y$, and the unique common neighbour c of b, y belongs to X: it cannot be v, and it cannot lie in $N(v)$, since b and y have different parents there. Thus $x - b - c - y$ is a path in X. Pairs with a common parent have no length-two path in X, so X has diameter three and Theorem 3.1 applies. Among the nonprincipal adjacency eigenvalues, $(-1+\sqrt{4K-3})/2$ maximizes $|\theta+1|$; substitution gives

$$\lambda_{\min}(D(X)) = -\frac{5+\sqrt{4K-3}}{2}.$$

The threshold reduces to

$$(2K-7)^2 - (4K-3) = 4(K^2 - 8K + 13) > 0$$

for $K \geq 6$. The finite $K = 7$ instance is checked by `scripts/verify_wow284_38_40_42.py`. □

3. The regular score calculus

Theorem 3.1 (Diameter-three score formula)**.** *Let G be connected, k-regular, of girth at least five and diameter three, with adjacency matrix A and order n. Then*

$$D = 3J + (k-3)I - 2A - A^2,$$
$$D + kI = 3J + (2k-2)I - (A+I)^2.$$

The principal distance eigenvalue is $3n - k^2 - k - 3$, and a nonprincipal adjacency eigenvalue θ gives the distance eigenvalue

$$\mu(\theta) = k - 2 - (\theta+1)^2.$$

Consequently

$$\boxed{\Phi(G) = 2k - 2 - \max_{\theta\neq k}(\theta+1)^2.}$$

Thus G is a strict counterexample exactly when

$$|\theta+1| < \sqrt{2k-2} \qquad (\theta \neq k).$$

Proof. Girth at least five gives the distance-two matrix $A_2 = A^2 - kI$, and diameter three gives $A_3 = J - I - A - A_2$. Substitute in $D = A + 2A_2 + 3A_3$. On $\mathbf{1}^{\perp}$, J vanishes, and regularity gives $\delta^* = k$. The principal distance eigenvalue is positive and is the Perron root because every off-diagonal entry of D is positive. The exact operator and score identities are independently checked by `scripts/verify_regular_score_calculus.py`. □

Corollary 3.2 (Bipartite obstruction)**.** *A connected k-regular bipartite graph of girth at least five and diameter three is not a strict counterexample for $k \geq 3$.*

Proof. The nonprincipal eigenvalue $-k$ violates the open shifted window, with equality only when $k = 3$. □

Proposition 3.3 (Higher-diameter transfer)**.** *Let G be connected and k-regular, with diameter d and girth at least $2d-1$. Let A_i denote the distance-i matrix, and define*

$$F_0 = 1, \quad F_1 = x, \quad F_2 = x^2 - k, \quad F_i = xF_{i-1} - (k-1)F_{i-2} \quad (i \geq 3).$$

Then $A_i = F_i(A)$ for $0 \leq i \leq d-1$, and

$$D = dJ + q_d(A), \qquad q_d(x) = \sum_{i=0}^{d-1}(i-d)F_i(x).$$

In particular,

$$q_3(x) = k - 3 - 2x - x^2,$$
$$q_4(x) = -x^3 - 2x^2 + (2k-4)x + 2k - 4.$$

Proof. Up to length $d-1$, the girth condition makes nonbacktracking walks between two vertices unique exactly when their length is the graph distance. Hence the distance-i matrices are the nonbacktracking polynomials in A; summing $D=\sum_{i=0}^{d} iA_i$ and eliminating A_d with $J=\sum_{i=0}^{d} A_i$ proves the formula. This lies within the established distance-polynomial framework [3, 4]. □

4. Degree and diameter obstructions

Lemma 4.1. *For every connected graph,*

$$\lambda_{\min}(D(G)) \le -\operatorname{diam}(G).$$

Proof. For a diametral pair u, v, the Rayleigh quotient of $e_u - e_v$ is $-d_G(u,v)$. □

Theorem 4.2. *Every connected regular strict counterexample to WOW-284 has degree at least six.*

Proof. We use the LP ceiling proved independently in Theorem 5.3; that theorem does not depend on the present degree reduction. Let the degree be k. Lemma 4.1 and strictness give $\operatorname{diam}(G) < k$. The cases $k \le 2$ are immediate. For $k=3$, the radius-two lower bound and diameter at most two force equality in the Moore bound. Hence the graph is a degree-three Moore graph, and Theorem 2.2 gives equality rather than strict violation.

For $k=4$, diameter two would require a degree-four Moore graph, whose adjacency multiplicities are nonintegral. In diameter three, the exact LP bound of Theorem 5.3 gives $n<19$, whereas the radius-two ball has 17 vertices and diameter three requires at least one more. Hence $n=18$. The distance-three matrix is then a perfect matching. Its -1-eigenspace W is a nine-dimensional rational subspace of $\mathbf{1}^{\perp}$, and

$$A_3 = J + 3I - A - A^2 \quad\Longrightarrow\quad (A^2 + A - 4I)|_W = 0.$$

The polynomial x^2+x-4 is irreducible over $\mathbb{Q}$, so a rational space on which it annihilates an operator has even dimension, a contradiction.

For $k=5$, diameter two again fails the Moore multiplicity condition. A diametral geodesic in diameter four yields the principal submatrix $D(P_5)$, whose factor x^2+6x+4 supplies the eigenvalue $-3-\sqrt{5} < -5$; Cauchy interlacing excludes this case. In diameter three, Meringer's lower bound and Theorem 5.3 leave $n \in \{30, 31, 32\}$, and parity removes 31. At $n=32$, write a for the average internal degree of the distance-two layer. This layer has 20 vertices, each with one neighbour in the first layer, so the number of edges from the second to the six-vertex third layer is $20(4-a) \le 6 \cdot 5$; hence $a \ge 5/2$. In the symmetric normalized layer compression, the derivative of the only a-dependent block is

$$\begin{pmatrix} 1 & -\sqrt{r} \\ -\sqrt{r} & r \end{pmatrix} = \begin{pmatrix} 1 \\ -\sqrt{r} \end{pmatrix} \begin{pmatrix} 1 & -\sqrt{r} \end{pmatrix} \succeq 0, \qquad r = \frac{5\cdot 4}{6}.$$

Thus every ordered compression eigenvalue is nondecreasing in a. At the smallest feasible value $a=5/2$, the relevant nonprincipal factor is

$$p_{5,6}(x) = 4x^3 + 10x^2 - 16x - 30, \qquad p_{5,6}(11/6) = -29/27 < 0.$$

Because the leading coefficient is positive, its largest root exceeds $11/6 > -1+\sqrt{8}$, contradicting the open shifted window. At $n=30$, Meringer's isomorph-free enumeration leaves exactly four $(5,5)$-cages [13]; each fixed record has an exact distance eigenvalue at most -5. The complete case split, the four fixed graph6 records, characteristic polynomials, and interlacing directions are independently checked by `scripts/verify_proof_audit_04_regular_low_degree.py`. □

Theorem 4.3 (Endpoint-neighbourhood obstruction)**.** *Let G be any connected finite simple graph, and let u, v be vertices at distance $\ell = d_G(u,v) \ge 5$. Put $p=d(u)$ and $q=d(v)$. Then*

$$\boxed{\lambda_{\min}(D(G)) \le p + q - 2 - \sqrt{(p-q)^2 + pq(\ell-2)^2}.}$$

If δ is the ordinary minimum degree, then

$$\boxed{\lambda_{\min}(D(G)) \le -\delta(\ell-4) - 2.}$$

Consequently every strict WOW-284 counterexample satisfies

$$\Delta > \delta(\ell-4) + 2,$$

where Δ is the ordinary maximum degree. In particular, every regular strict counterexample has diameter at most four.

Proof. The two endpoint neighbourhoods are disjoint. Give weight $a > 0$ to $N(u)$, weight $-b < 0$ to $N(v)$, and zero elsewhere. Within one neighbourhood distances are at most two; between the two neighbourhoods they are at least $\ell - 2$. Since the cross products are negative, the Rayleigh quotient is at most that of

$$\begin{pmatrix} 2(p-1) & -(\ell-2)\sqrt{pq} \\ -(\ell-2)\sqrt{pq} & 2(q-1) \end{pmatrix}.$$

Its least eigenvalue is the first displayed bound, and its least eigenvector can be chosen with both coordinates positive. Write $p = \delta + \alpha$, $q = \delta + \beta$, and $t = \ell - 2$. The identity

$$(p-q)^2 + pqt^2 - (p+q+\delta(t-2))^2 = (t-2)\{\delta t(\alpha+\beta) + (t+2)\alpha\beta\}$$

is nonnegative and gives the second bound. Finally $\delta^*(G) \leq \Delta$. The sign choice, radical comparison, and integer rounding are independently audited by `scripts/verify_proof_audit_10_endpoint_diameter.py`. □

Theorem 4.4 (Diameter four)**.** *Let G be connected, k-regular, of girth at least five and diameter four. Then*

$$\boxed{\lambda_{\min}(D(G)) \leq -\frac{7+\sqrt{16k+1}}{2}.}$$

Consequently no such graph of degree $2 \leq k \leq 9$ is a strict counterexample.

Proof. Choose u, v at distance four and put $U = N(u)$, $V = N(v)$. For fixed $a \in U$, distinct vertices $b, b' \in V$ at distance two from a cannot use the same common neighbour, or a 4-cycle results. Thus at most $k(k-1)$ pairs in $U \times V$ have distance two, and

$$\sum_{a \in U, b \in V} d_G(a,b) \geq 2k^2 + k.$$

Assign weights $\alpha, \beta, -\alpha, -\beta$ to u, U, v, V, respectively. Counting unordered pairs and then doubling gives

$$\frac{x^{\mathsf{T}} D(G) x}{x^{\mathsf{T}} x} \leq \frac{-4\alpha^2 - 4k\alpha\beta - 3k\beta^2}{\alpha^2 + k\beta^2}.$$

After setting $y_1 = \alpha$ and $y_2 = \sqrt{k}\,\beta$, the right-hand side is the Rayleigh quotient of

$$\begin{pmatrix} -4 & -2\sqrt{k} \\ -2\sqrt{k} & -3 \end{pmatrix},$$

whose least eigenvalue is the displayed value and has a positive-coordinate minimizer. The strict comparison with $-k$ holds for $2 \leq k \leq 9$. Every orientation factor, cross-distance sign, and endpoint comparison is independently checked by `scripts/verify_proof_audit_11_diameter_four.py`. □

Corollary 4.5 (Regular trichotomy)**.** *Every regular strict counterexample has one of the following forms:*

(1) diameter two, hence a Moore graph;
(2) diameter three, with all nonprincipal adjacency eigenvalues in the open shifted WOW window;
(3) diameter four, with degree at least ten.

There are no regular strict counterexamples of diameter at least five.

5. Moment bounds and the exact LP ceiling

Let G satisfy the hypotheses of Theorem 3.1, and write its nonprincipal adjacency eigenvalues as θ_i. Put $y_i = \theta_i + 1$.

Proposition 5.1 (Fourth-moment identity)**.** *One has*

$$\sum_{i=1}^{n-1}(2k-2-y_i^2)(y_i+1)^2 = (k+2)\big((k+2)(k^2+3)-6n\big).$$

Every strict counterexample therefore satisfies

$$\boxed{n < B_k := \frac{(k+2)(k^2+3)}{6}.}$$

Proof. Use

$$\operatorname{tr} A = \operatorname{tr} A^3 = 0, \quad \operatorname{tr} A^2 = nk, \quad \operatorname{tr} A^4 = nk(2k-1),$$

remove the principal eigenvalue, and expand both sides. In a strict counterexample, each factor $2k-2-y_i^2$ is positive. The sum cannot vanish: otherwise every nonprincipal adjacency eigenvalue would equal -2, and $\operatorname{tr} A = 0$ would give $k-2(n-1) = 0$, or $n = (k+2)/2$, incompatible with the elementary bound $n \geq k+1$ for a simple k-regular graph. The identity is checked symbolically by `scripts/verify_degree_six_gate.py` and independently within `scripts/verify_proof_audit_02_two_sided_lp.py`. □

The preceding bound is in fact optimal within the entire standard one-variable nonbacktracking LP hierarchy.

Definition 5.2. Let $F_i = F_i^{(k)}$ be the nonbacktracking polynomials from Proposition 3.3, and set

$$I_k = [-1-\sqrt{2k-2}, -1+\sqrt{2k-2}].$$

A finite polynomial $f = \sum_i f_i F_i$ is *admissible* if

$$f_0 > 0, \qquad f_i \geq 0 \quad (i \geq 5), \qquad f(x) \leq 0 \quad (x \in I_k).$$

For a k-regular graph of girth at least five whose nonprincipal spectrum lies in I_k, one has $\operatorname{tr} F_i(A) = 0$ for $1 \leq i \leq 4$, while $\operatorname{tr} F_i(A) \geq 0$ for $i \geq 5$, since these traces count closed nonbacktracking walks. The coefficient conditions therefore give $nf_0 \leq \operatorname{tr} f(A)$. On the other hand, $f(\theta) \leq 0$ for every nonprincipal eigenvalue, so $\operatorname{tr} f(A) \leq f(k)$. Thus

$$nf_0 \leq \operatorname{tr} f(A) \leq f(k), \qquad n \leq \frac{f(k)}{f_0}.$$

Theorem 5.3 (Exact LP ceiling and rigidity)**.** *For every integer $k \geq 4$ and every admissible f,*

$$\boxed{\frac{f(k)}{f_0} \geq B_k = \frac{(k+2)(k^2+3)}{6}.}$$

Equality holds if and only if f is a positive scalar multiple of

$$\boxed{f_*(x) = \frac{(x+2)^2(x^2+2x-(2k-3))}{6(k+2)}.}$$

Thus increasing the polynomial degree cannot improve this one-variable LP bound. Consequently, any connected k-regular graph of girth at least five whose nonprincipal spectrum lies in the interior of I_k satisfies $n < B_k$.

Proof. The primal expansion is

$$\begin{aligned} 6(k+2)f_*(x) &= 6(k+2)F_0(x) + 2(2k+7)F_1(x) \\ &\quad + (k+13)F_2(x) + 6F_3(x) + F_4(x), \end{aligned}$$

On I_k, the factor $(x+1)^2-(2k-2)$ is nonpositive, while $(x+2)^2 \geq 0$; hence f_* is admissible and $f_*(k) = B_k$.

For the dual certificate, put $\Delta = \sqrt{2k-2}$, $\xi_\pm = -1 \pm \Delta$, and $\xi_0 = -2$. Define

$$w_- = \frac{k(k+2)(2k^2-6-3(k-1)\Delta)}{24(2k-3)},$$
$$w_0 = \frac{k(k-1)(k^2+3)}{6(2k-3)},$$
$$w_+ = \frac{k(k+2)(2k^2-6+3(k-1)\Delta)}{24(2k-3)}.$$

All weights are positive. For the only nontrivial one, this follows from

$$(2k^2-6)^2 - 18(k-1)^3 = 2(k-3)(2k-3)(k^2+3) > 0.$$

The measure $\mu = w_-\delta_{\xi_-} + w_0\delta_{\xi_0} + w_+\delta_{\xi_+}$ satisfies

$$\mu(1) = B_k - 1, \qquad \mu(F_i) = -F_i(k) \quad (1 \le i \le 4).$$

For $i \ge 5$, the slack $a_i = \mu(F_i) + F_i(k)$ is strictly positive. For $5 \le i \le 9$, exact calculation gives, after removing the common positive factor $k(k-1)(k+2)(k^2+3)/6$, respectively,

$$2, \quad 5k-13, \quad 2(3k^2-17k+25),$$
$$6k^3-47k^2+139k-150, \quad 2(3k^4-27k^3+106k^2-219k+194),$$

all positive for $k \ge 4$; for the nontrivial residual factors this follows after writing $k = m+4$, when all coefficients are nonnegative and the constant terms are positive. For $i \ge 10$, put $r = k-1 \ge 3$. The support lies in $[-2\sqrt{r}, 2\sqrt{r}]$, because $1+\sqrt{2r} \le 2\sqrt{r}$. For $|z| \le 1$, the recurrence gives

$$F_i(2\sqrt{r}\,z) = r^{i/2}U_i(z) - r^{(i-2)/2}U_{i-2}(z),$$

where U_j is the Chebyshev polynomial of the second kind. Using $|U_j(z)| \le j+1$ yields

$$\frac{|\mu(F_i)|}{F_i(k)} \le \frac{2i+1}{3}\,3^{3-i/2}.$$

At $i = 10$ the right-hand side is $7/9$, and it decreases thereafter because $3(2i+1)^2 - (2i+3)^2 = 8i^2 - 6 > 0$. Thus it is strictly less than one for every $i \ge 10$. Expanding in the nonbacktracking basis gives

$$\int f\,d\mu = B_k f_0 - f(k) + \sum_{i\ge5} f_i a_i \ge B_k f_0 - f(k).$$

Since $f \le 0$ on the support of μ, the left-hand side is at most zero, which proves weak duality. If equality holds, strict positivity of every high-degree slack forces $f_i = 0$ for $i \ge 5$. Equality on the positive dual support forces zeros at $\xi_-, -2, \xi_+$; the interior zero -2 has even multiplicity because $f \le 0$ on I_k. Degree at most four then forces f to be a scalar multiple of f_*, and $f_0 > 0$ makes the scalar positive. Equality in the graph bound would then force every nonprincipal adjacency eigenvalue to be -2, which contradicts the trace equation; hence the open-window bound is strict. Every symbolic identity, finite slack, tail bound, and equality-nullspace calculation is independently checked by `scripts/verify_proof_audit_02_two_sided_lp.py`. □

6. Edge-local positivity and the degree-six boundary

Theorem 5.3 gives $n \le 51$ for a degree-six regular strict counterexample. The last order is eliminated by using local cycle data that the one-point LP trace bound does not see.

Proposition 6.1 (Edge-local cycle bounds)**.** *Let G be connected and k-regular, of girth at least five and diameter three, and suppose its nonprincipal adjacency spectrum lies in the closed shifted WOW window. For an edge uv, let σ_{uv} be the number of 5-cycles containing that edge. Then*

$$2k-2 \le \sigma_{uv} \le \frac{2(k+2)^2(k^2+3)}{n} - 10k - 26.$$

If $n = k^2+1+c$, then also

$$\sigma_{uv} \ge (k-1)^2 - c.$$

Proof. Put

$$f_k(x) = (x+2)^2((x+1)^2 - (2k-2)), \quad C_k = f_k(k),$$

and

$$M = -f_k(A) + \frac{C_k}{n} J.$$

The spectral window gives $M \succeq 0$. For an edge uv,

$$(A^3)_{uv} = \sum_{z \sim v} (A^2)_{uz} = k + (k-1) = 2k - 1.$$

Here $z = u$ contributes k, while every other neighbour of v is at distance two from u and has a unique length-two path from u. Moreover,

$$(A^4)_{uv} = \sum_z (A^2)_{uz}(A^2)_{zv} = \sigma_{uv}.$$

Indeed, the nonzero summands away from the diagonal are precisely the vertices at distance two from both u and v. Their two unique length-two paths, together with uv, form a five-cycle, and each five-cycle through uv yields one such vertex. Since $C_k = (k+2)^2(k^2+3)$, the diagonal and edge entries of M are

$$a = \frac{C_k}{n} - 6(k+2), \qquad b = \frac{C_k}{n} - (4k+14) - \sigma_{uv}.$$

The principal submatrix on $\{u, v\}$ is $\left(\begin{smallmatrix} a & b \\ b & a \end{smallmatrix}\right)$, so $a \geq 0$ and $-a \leq b \leq a$. The inequality $b \leq a$ gives $\sigma_{uv} \geq 2k-2$, while $b \geq -a$ gives the stated upper bound.

For the final bound, every radius-two ball has size $k^2 + 1$. The set

$$\{u, v\} \cup (N(u) \setminus \{v\}) \cup (N(v) \setminus \{u\})$$

contains $2k$ vertices and lies in $B_2(u) \cap B_2(v)$. Every further intersection vertex is at distance two from both endpoints and is therefore in the preceding five-cycle bijection. Hence

$$|B_2(u) \cap B_2(v)| = 2k + \sigma_{uv}.$$

Inclusion–exclusion and $n = k^2 + 1 + c$ now give $\sigma_{uv} \geq (k-1)^2 - c$. The complete walk classification, sign directions, and radius-two bijection are independently checked by `scripts/verify_proof_audit_01_edge_local.py`. □

Corollary 6.2 (Edge–cycle divisibility sieve)**.** *Under the hypotheses of Proposition 6.1, put*

$$L_{k,n} = \max\{2k-2,\ 2k^2 - 2k + 2 - n\}, \qquad U_{k,n} = \frac{2(k+2)^2(k^2+3)}{n} - 10k - 26.$$

Necessarily

$$\boxed{\lceil L_{k,n} \rceil \leq \lfloor U_{k,n} \rfloor.}$$

If both sides equal an integer s, then every edge of G lies in exactly s five-cycles and

$$\boxed{5 \mid \frac{skn}{2}.}$$

Proof. Writing $n = k^2 + 1 + c$, the two lower bounds in Proposition 6.1 combine to

$$\sigma_{uv} \geq \max\{2k-2, (k-1)^2 - c\} = L_{k,n},$$

while its upper bound is $U_{k,n}$. Since σ_{uv} is an integer, the first conclusion follows. If the two integer bounds coincide at s, then $\sigma_{uv} = s$ for every edge. Counting edge–five-cycle incidences gives

$$5N_5 = \sum_{uv \in E(G)} \sigma_{uv} = s|E(G)| = \frac{skn}{2},$$

which proves the divisibility condition. □

Theorem 6.3. *Every connected* 6*-regular strict counterexample to WOW-284 has order at most* 50.

Proof. A separate Rayleigh and trace argument shows that any degree-six strict counterexample has diameter three. For vertices at distance $d \geq 4$, the vector with weights $3, 1, -3, -1$ on the two endpoints and their respective neighbourhoods has Rayleigh quotient at most

$$\frac{204 - 81d}{15} \leq -8,$$

contradicting $\delta^* = 6$. Diameter two would force equality in the Moore bound, hence order 37 and adjacency characteristic polynomial $(x-6)(x^2+x-5)^{18}$; its root sum is -12, contradicting $\operatorname{tr} A = 0$. The LP ceiling gives $n \leq 51$.

Assume $n = 51$. Corollary 6.2 has $\lceil L_{6,51} \rceil = \lfloor U_{6,51} \rfloor = 11$, and hence requires

$$5N_5 = 153 \cdot 11 = 1683,$$

impossible modulo five. The full diameter reduction and incidence audit are contained in `scripts/verify_proof_audit_01_edge_local.py`. □

Corollary 6.4 (Low-degree order windows)**.** *The current bounds imply:*

$$\begin{aligned} k = 6: &\quad n \leq 50, \\ k = 7: &\quad n = 50 \textit{ in diameter two, or } n \leq 76 \textit{ in diameter three}, \\ k = 8: &\quad n \leq 110, \\ k = 9: &\quad n \leq 152. \end{aligned}$$

Proof. Theorem 4.4 removes diameter four for these degrees. At $k = 7$, the diameter-two case is the order 50 Hoffman–Singleton graph, whereas Theorem 5.3 gives $n < 78$ in diameter three; the handshake lemma makes n even, so $n \leq 76$. At $k = 8$ and $k = 9$, the Moore multiplicities are nonintegral, so diameter two is impossible. The LP ceiling gives $n \leq 111$ for $k = 8$ and $n \leq 153$ for $k = 9$; parity improves the latter to 152. Finally, at $k = 8, n = 111$, Corollary 6.2 has $\lceil L_{8,111} \rceil = \lfloor U_{8,111} \rfloor = 14$, but it would require

$$5N_5 = 14 \cdot 444 = 6216$$

is impossible. Exact integer arithmetic is checked by `scripts/verify_low_degree_windows.py`. □

6.1. **Necessary structure at order fifty.** The remaining degree-six boundary is highly constrained, although not yet eliminated.

Theorem 6.5. *Let G be a connected* 6*-regular graph of order* 50 *and girth at least five, and suppose* $\Phi(G) > 0$*. Then G has diameter three. Every edge lies in* 12 *or* 13 *five-cycles. Let H be the spanning subgraph of edges of the second type, let $m = |E(H)|$, and let $\tau(v)$ be the number of five-cycles through v. Then*

$$\tau(v) \in \{36, 37, 38\}, \qquad d_H(v) = 2\tau(v) - 72 \in \{0, 2, 4\},$$
$$m \equiv 0 \pmod 5, \qquad N_5 = 360 + \frac{m}{5}.$$

For a two-edge path $u - v - w$, put

$$r_{uvw} = 6\alpha_{uvw} + \beta_{uvw},$$

where α and β count the five- and six-cycles containing the path. The allowed values are

types of the two incident edges	r_{uvw}
low–low	30, 31, 32
mixed	30, 31, 32
high–high	30, 31.

Writing $S_2 = \sum_v d_H(v)^2$, one has

$$1950 - m \leq N_6 \leq 2200 - \frac{5m}{6} - \frac{S_2}{12},$$

$$\frac{43m^2 - 70200m + 119632500}{58500} \leq N_6 \leq \frac{4220000 - 2200m - 7m^2}{2000}.$$

Exact enumeration of the resulting coarse degree profiles leaves 266 *possibilities.*

Proof. The diameter reduction used in Theorem 6.3 applies verbatim: diameter at least four gives a Rayleigh quotient at most -8, and diameter two gives the trace contradiction from $(x-6)(x^2+x-5)^{18}$. Hence the diameter is three. Around a fixed vertex the distance layers have sizes $1, 6, 30, 13$. If τ is the number of five-cycles through the centre, the average row quotient is similar to the symmetric compression on normalized layer indicators. Its nonprincipal factor q_τ satisfies

$$195q_\tau(-1+\sqrt{10}) = (-215+56\sqrt{10})\tau + 7350 - 1860\sqrt{10}.$$

The coefficient of τ is negative because $56^2 \cdot 10 < 215^2$, and at $\tau = 39$ the right-hand side is $9(-115+36\sqrt{10}) < 0$. Also

$$195q_\tau(6) = 1500(75-\tau) > 0.$$

The last inequality is strict: the third layer is nonempty and connected to the second, while their edge count is $150-2\tau$. Thus $\tau \geq 39$ would place a nonprincipal compression eigenvalue in $(-1+\sqrt{10}, 6)$, contradicting interlacing and the open WOW window. Conversely, $150 - 2\tau \leq 13 \cdot 6$, so $\tau \geq 36$. Hence $\tau \in \{36, 37, 38\}$. Finally, $\sum_{e \ni v} \sigma_e = 2\tau(v)$, so the high-edge degree is $2\tau(v) - 72$; counting edge–five-cycle incidences gives $5N_5 = 12 \cdot 150 + m$.

For a two-path $u - v - w$, girth at least five gives $(A^3)_{uw} = \alpha_{uvw}$ and $(A^4)_{uw} = 16 + \beta_{uvw}$. The corresponding 3×3 principal minor of the centered positive-semidefinite matrix yields the displayed finite sets for r_{uvw}, except for an apparent equality value $r = 29$. At equality, the Gram norm of $e_u - e_w$ vanishes, so this vector lies in the kernel of the centered matrix. On $\mathbf{1}^\perp$, that matrix is $-f_6(A)$. The strict shifted window excludes the two endpoint zeros of f_6, leaving only its double zero at -2; hence the kernel there is exactly the adjacency -2-eigenspace. However, the u-coordinate of $A(e_u - e_w)$ is zero because $u \nsim w$, whereas the u-coordinate of $-2(e_u - e_w)$ is -2. This excludes $r = 29$. Summing the local inequalities gives the first pair of N_6 bounds; shifted moment and localizing matrices give the second. The complete symbolic determinants, Schur complements, kernel argument, and integer enumeration are independently checked by `scripts/verify_proof_audit_06_order50_feasibility.py`. The surviving 266 profiles are exact pruning data, not an existence or nonexistence claim. □

7. Distance spectra of punctured Moore graphs

Let M be a degree-k Moore graph of diameter two and put $\Delta = \sqrt{4k-3}$.

Theorem 7.1 (One deleted vertex)**.** *For $H = M - v$,*

$$|V(H)| = k^2, \qquad \delta^*(H) = k - \frac{1}{k}, \qquad \lambda_{\min}(D(H)) = -2 - \sqrt{k}.$$

The complete distance spectrum is

$$\operatorname{Spec} D(H) = \{\rho_+, \rho_-\} \cup \{(-2+\sqrt{k})^{(k-1)}, (-2-\sqrt{k})^{(k-1)}\}$$
$$\cup \left\{ \left(-\frac{\Delta+3}{2}\right)^{(m_+)}, \left(\frac{\Delta-3}{2}\right)^{(m_-)} \right\},$$

where

$$\rho_\pm = k^2 - 2 \pm \sqrt{k(k^3 - 2k^2 + 3k - 1)},$$
$$m_\pm = \frac{k(k-2)(\Delta \pm 1)}{2\Delta}.$$

Thus

$$\Phi(H) = k - \frac{1}{k} - 2 - \sqrt{k},$$

which is positive exactly for integers $k \geq 5$.

Theorem 7.2 (Endpoints of an edge)**.** *Let $uv \in E(M)$ and $H = M - \{u, v\}$. For $k \geq 3$,*

$$|V(H)| = k^2 - 1, \qquad \delta^*(H) = k - \frac{2}{k}, \qquad \lambda_{\min}(D(H)) = -2 - \sqrt{k}.$$

The complete distance spectrum is

$$\begin{aligned}\operatorname{Spec} D(H) = \{&\sigma_+, \sigma_-, k-4\} \\ &\cup \{(-2+\sqrt{k})^{(2k-4)}, (-2-\sqrt{k})^{(2k-4)}\} \\ &\cup \left\{\left(-\frac{\Delta+3}{2}\right)^{(a_+)}, \left(\frac{\Delta-3}{2}\right)^{(a_-)}\right\},\end{aligned}$$

where

$$\sigma_\pm = k^2 - 3 \pm \sqrt{k^4 - 2k^3 + 3k^2 - 8k + 7},$$
$$a_+ = \frac{(k-2)(k+(k-2)\Delta)}{2\Delta}, \quad a_- = \frac{(k-2)((k-2)\Delta - k)}{2\Delta}.$$

Thus

$$\Phi(H) = k - \frac{2}{k} - 2 - \sqrt{k},$$

which is positive exactly for integers $k \geq 5$.

Proof architecture for Theorems 7.1 and 7.2. The displayed dual-degree formulas are the cases $s = 1$ and $s = 2$ of Theorem 8.1; its proof below is independent of the spectral decompositions. For one deleted vertex, put $A = N(v)$, $B = \Gamma_2(v)$, let C be the A-by-B incidence matrix, and let B_0 be the adjacency matrix on B. Only pairs inside A lose their unique length-two path. For distinct $a, a' \in A$, choose a neighbour $b \in B$ of a. The vertices b, a' are nonadjacent, and their unique common neighbour c lies in B: it is not v, since $b \nsim v$, and it is not in A, since an edge inside A would form a triangle through v. Hence $a - b - c - a'$ is a surviving path, and therefore

$$D(H) = \begin{pmatrix} 3(J-I) & 2J - C \\ 2J - C^\mathsf{T} & 2(J-I) - B_0 \end{pmatrix}.$$

The normalized constant quotient is

$$Q_1 = \begin{pmatrix} 3(k-1) & (2k-1)\sqrt{k-1} \\ (2k-1)\sqrt{k-1} & 2k^2 - 3k - 1 \end{pmatrix},$$

whose eigenvalues are $\rho_\pm$. The identities $CC^\mathsf{T} = (k-1)I$, $CB_0 = J - C$, and the bottom-right block of the Moore relation yield

$$\mathbb{R}^{V(H)} = W_{\mathrm{const}} \perp W_{\mathrm{inc}} \perp \ker C, \qquad 2 + 2(k-1) + k(k-2) = k^2.$$

On each zero-sum incidence module the action matrix is $\left(\begin{smallmatrix} -3 & -(k-1) \\ -1 & -1 \end{smallmatrix}\right)$, giving $-2 \pm \sqrt{k}$. On $\ker C$, $B_0^2 + B_0 - (k-1)I = 0$, $D = -2I - B_0$, and the trace of B_0 is zero after removing the constant and incidence images; dimension and trace give $m_\pm$.

For an adjacent deleted pair, put $A = N(u) \setminus \{v\}$, $B = N(v) \setminus \{u\}$, and let C be the residual cell. Only pairs inside A or inside B lose a length-two path. For distinct $a, a' \in A$, choose a residual neighbour c of a. The vertices c, a' are nonadjacent, and their unique common neighbour is also residual: it cannot be one of the deleted endpoints, cannot lie in A by triangle-freeness, and cannot lie in B because no edge joins A to B. This gives a surviving length-three path; the argument for B is symmetric. The antisymmetric constant line has eigenvalue $k - 4$, and the normalized symmetric quotient is

$$Q_2 = \begin{pmatrix} 5k - 8 & \sqrt{(k-1)(2k-3)(4k-6)} \\ \sqrt{(k-1)(2k-3)(4k-6)} & 2k^2 - 5k + 2 \end{pmatrix},$$

with eigenvalues $\sigma_\pm$. Two zero-sum incidence modules and a residual kernel complete the decomposition, with

$$3 + 2(2k-4) + (k-2)^2 = k^2 - 1.$$

The residual adjacency trace is $k - 2$, giving $a_\pm$. The residual negative root is greater than $-2 - \sqrt{k}$. The same is true of the smaller constant-quotient roots: in each case the leading diagonal entry of the shifted quotient is positive, and

$$\det(Q_1 + (2+\sqrt{k})I) = k(2k^2 + 2k^{3/2} - 3k + 2) > 0,$$
$$\det(Q_2 + (2+\sqrt{k})I) = (\sqrt{k}-1)(\sqrt{k}+1)(2k^2 + 2k^{3/2} - 3k + 2\sqrt{k} + 6) > 0.$$

Exact quotient normalization, trace-to-multiplicity equations, replacement paths, and all least-root comparisons are independently checked by `scripts/verify_proof_audit_05_small_moore_punctures.py`. □

Theorem 7.3 (Two nonadjacent deleted vertices)**.** *Let u, v be nonadjacent vertices of M, $k \geq 5$, and put $H = M - \{u, v\}$. Define*

$$\begin{aligned} R_k(x) = x^4 &+ (10 - 2k^2)x^3 + (2k^3 - 17k^2 - 2k + 36)x^2 \\ &+ (12k^3 - 49k^2 - 4k + 53)x \\ &- 2k^4 + 17k^3 - 38k^2 + 5k + 20, \end{aligned}$$

$$M_- = \frac{k(k-2) + (k^2 - 4k + 2)\Delta}{2\Delta}, \quad M_+ = \frac{-k(k-2) + (k^2 - 4k + 2)\Delta}{2\Delta}.$$

Then

$$\begin{aligned} \chi_{D(H)}(x) = (x - k + 3)&R_k(x)(x^2 + 4x - k + 3)^{k-2} \\ &\cdot (x^2 + 4x - k + 5)^{k-2}\left(x + \frac{\Delta + 3}{2}\right)^{M_-} \\ &\cdot \left(x - \frac{\Delta - 3}{2}\right)^{M_+}. \end{aligned}$$

Moreover,

$$\delta^*(H) = k - \frac{2}{k},$$

and H is a strict counterexample for every realizable integer $k \geq 6$.

Proof. The deleted vertices have a unique common neighbour w. With $A = N(u) \setminus \{w\}$, $B = N(v) \setminus \{w\}$, $C = N(w) \setminus \{u, v\}$, and residual cell Z, the surviving graph has an equitable five-cell geometry and a perfect matching between A and B. Explicit length-three paths show that every pair whose unique length-two path was deleted has new distance exactly three. The five-cell distance quotient has characteristic polynomial $(x - k + 3)R_k(x)$.

Let R_A, R_B, R_C be the incidence matrices from the three nonconstant cells to Z, and let T be the adjacency matrix on Z. The Moore identity splits the remaining space into matched symmetric and antisymmetric modules, a common-neighbour module, and $K = \ker R_A \cap \ker R_B \cap \ker R_C$. The first three modules give the two displayed quadratic factors and $k - 3$ copies of each Moore linear factor. On K,

$$T^2 + T - (k-1)I = 0, \qquad D = -2I - T.$$

The constant direction of Z has T-eigenvalue $k - 3$, the symmetric, antisymmetric, and common-neighbour images have eigenvalues $-2, 0, -1$ with dimensions $k-2, k-2, k-3$. Since T has zero diagonal, its total trace is zero, and therefore $\mathrm{tr}(T|_K) = 2(k-2)$. Dimension and trace now give the residual multiplicities; after adding the common-neighbour copies they are precisely M_- and M_+. The full dimension count is

$$5 + 4(k-2) + 2(k-3) + (k-2)(k-4) = k^2 - 1.$$

The value $\delta^*(H) = k - 2/k$ is again the $s = 2$ case of Theorem 8.1. For strictness, the distance-increase matrix is the adjacency matrix of two copies of K_k meeting in w, and hence has least eigenvalue

$$\lambda_{\min}(E_S) = \frac{k - 2 - \sqrt{k^2 + 4k - 4}}{2}.$$

Since the parent Moore graph has score $k - (3 + \Delta)/2$ and deletion lowers the minimum dual degree by $2/k$, Proposition 7.4 gives

$$\Phi(H) \geq \frac{3k - 5 - \Delta - \sqrt{k^2 + 4k - 4}}{2} - \frac{2}{k}.$$

For $k \geq 6$, the sign-preserving squarings reduce positivity of this lower bound to $P(k) > 0$, where

$$P(k) = 4k^8 - 39k^7 + 95k^6 + 9k^5 - 173k^4 - 36k^3 + 116k^2 + 80k + 16.$$

Writing $k = m + 6$ gives

$$\begin{aligned} P(m+6) &= 4m^8 + 153m^7 + 2489m^6 + 22329m^5 \\ &\quad + 119437m^4 + 382236m^3 + 685268m^2 \\ &\quad + 559616m + 75952 > 0. \end{aligned}$$

The direct-sum decomposition, injectivity, orthogonality, all recomputed distances, and the polynomial sign are independently checked by `scripts/verify_proof_audit_03_nonadjacent_puncture.py` and `scripts/verify_research_extensions_exact.py`. □

Proposition 7.4 (Deletion stability). *Let G be connected, $S \subseteq V(G)$, and suppose $H = G - S$ is connected. Put*

$$D_0 = D(G)[V(H)], \qquad E_S = D(H) - D_0,$$
$$a = \delta^*(G), \quad b = \delta^*(H), \quad \gamma = \Phi(G).$$

Then

$$\boxed{\Phi(H) \geq \gamma - (a-b) + \lambda_{\min}(E_S).}$$

Proof. One has

$$D(H) + bI = (D_0 + aI) + E_S - (a-b)I.$$

Principal-submatrix interlacing gives $\lambda_{\min}(D_0 + aI) \geq \gamma$, and Weyl's inequality gives the result. For Moore punctures, E_S is an explicit distance-increase graph; the exact specializations are checked by `scripts/verify_research_extensions_exact.py`. □

8. Small punctures and exact Hoffman–Singleton robustness

Theorem 8.1 (Small-puncture normal form). *Let M be a degree-k Moore graph, let $S \subseteq V(M)$ have size $s \leq k-1$, and put $H = M - S$. Then H is connected, has diameter at most three, and*

$$\boxed{\delta^*(H) = k - \frac{s}{k}.}$$

Let B be the surviving-vertex by deleted-vertex incidence matrix, with $B_{xz} = 1$ exactly when $x \sim_M z$. Then

$$\boxed{D(H) = 2(J-I) - A(H) + BB^{\mathsf{T}} - \operatorname{diag}(BB^{\mathsf{T}}).}$$

Proof. The case $s = 0$ is immediate, so assume $1 \leq s \leq k-1$. Let $x, y \in V(H)$ be nonadjacent in M, and suppose their unique common neighbour z lies in S. For every $a \in N_M(x) \setminus \{z\}$, the vertices a, y are nonadjacent and have a unique common neighbour b_a. Thus

$$x - a - b_a - y$$

is a length-three path. These $k-1$ paths are internally vertex-disjoint: a shared vertex $b_a = b_{a'}$ would give a four-cycle, while $b_a = a'$ would give a triangle through x. Besides z, at most $s - 1 \leq k-2$ vertices are deleted, so one path survives. Hence a destroyed length-two path becomes distance exactly three, which proves the matrix formula.

For $x \in V(H)$, let $t_x = |N_M(x) \cap S|$. A deleted neighbour of x is adjacent to no surviving neighbour of x, by triangle-freeness; each other deleted vertex has at most one common neighbour with x. Thus

$$\sum_{y \in N_H(x)} t_y \leq s - t_x.$$

Consequently,

$$d_H^*(x) \geq k - \frac{s - t_x}{k - t_x} \geq k - \frac{s}{k},$$

where the last inequality follows from

$$\left(k - \frac{s-t}{k-t}\right) - \left(k - \frac{s}{k}\right) = \frac{t(k-s)}{k(k-t)} \geq 0.$$

The intersection bound

$$\left| \bigcap_{z \in S} \Gamma_2(z) \right| \geq k^2 + 1 - s(k+1) \geq 2$$

provides a surviving vertex x at distance two from every deleted vertex. For $z \in S$, let y_z be the unique common neighbour of x, z. If $y_z \in S$, then the choice of x would require $d_M(x, y_z) = 2$, contradicting $x \sim y_z$. Thus all witnesses survive, each deleted vertex contributes exactly once to the neighbour-degree deficit, and equality is attained. The replacement paths, boundary case $s = k - 1$, distance formula, and attainment step are independently checked by `scripts/verify_proof_audit_12_small_puncture.py`. □

Corollary 8.2 (Uniform deletion stability)**.** *Under the hypotheses of Theorem 8.1, put*

$$t_x = |N_M(x) \cap S|, \qquad \tau(S) = \max_{x \in V(M-S)} t_x.$$

Then

$$\boxed{\Phi(M - S) \geq k - \frac{3 + \sqrt{4k-3}}{2} - \frac{s}{k} - \tau(S).}$$

In particular, the deletion is a strict counterexample whenever the right-hand side is positive. Since $\tau(S) \leq s$, every deletion of s vertices is strict whenever

$$\boxed{s\left(1 + \frac{1}{k}\right) < k - \frac{3 + \sqrt{4k-3}}{2}.}$$

Proof. By Theorem 8.1, the distance-increase matrix is

$$E_S = BB^{\mathsf{T}} - \operatorname{diag}(t_x).$$

Since $BB^{\mathsf{T}} \succeq 0$ and $\operatorname{diag}(t_x) \preceq \tau(S) I$, one has $\lambda_{\min}(E_S) \geq -\tau(S)$. The parent Moore graph has score

$$\Phi(M) = k - \frac{3 + \sqrt{4k-3}}{2},$$

and Theorem 8.1 shows that deletion lowers the minimum dual degree by s/k. Proposition 7.4 now gives the first bound. The second follows from $\tau(S) \leq s$. □

Theorem 8.3 (Hoffman–Singleton robustness radius)**.** *Let M be the Hoffman–Singleton graph. Every induced graph $M - S$ with $|S| \leq 5$ is a strict counterexample to WOW-284. This is sharp in the universal sense: there exists a six-vertex set whose deletion is not strict. Hence the universal vertex-deletion robustness radius is exactly five.*

Proof. Theorem 8.1 gives

$$\delta^*(M - S) = \frac{49 - |S|}{7}.$$

Corollary 8.2 already proves strictness uniformly for $|S| \leq 2$. To obtain the sharp universal radius, two explicitly stored permutations are verified edge-by-edge as automorphisms of the coordinate graph; the group they generate has order 252000. Their orbits on deletion sets of sizes $0, 1, 2, 3, 4, 5$ have counts

$$1, 1, 2, 4, 11, 33.$$

For every one of the 52 representatives, exact fraction arithmetic gives

$$7D(M - S) + (49 - |S|)I \succ 0.$$

Orbit-size sums equal $\binom{50}{s}$, so every labelled set is covered.

For sharpness, delete

$$\{P_{2,4}, P_{3,1}, P_{3,4}, Q_{2,1}, Q_{3,4}, Q_{4,4}\}.$$

The resulting graph has $\delta^* = 43/7$, while an exact LDL^{T} decomposition of $7D + 43I$ has exactly one negative and no zero pivot. Thus it is not strict. Generator action, orbit exhaustion, BFS distances, the small-puncture formula, and a handwritten rational LDL^{T} implementation are independently checked by `scripts/verify_proof_audit_13_hs_robustness.py`. □

Remark 8.4. The theorem asserts that every deletion through size five succeeds and that at least one deletion of size six fails. It does not assert that every six-vertex deletion fails.

9. Equality and obstructions to natural construction families

Theorem 9.1 (Equality boundary)**.** *Let G be connected, k-regular, of girth at least five and diameter three. Then*

$$\Phi(G) = 0 \quad \Longleftrightarrow \quad \max_{\theta \neq k} |\theta + 1| = \sqrt{2k-2}.$$

Equivalently, $D + kI$ is positive semidefinite and singular. If $2k-2$ is not a square, the two boundary adjacency eigenvalues occur with equal multiplicity, so the distance eigenvalue $-k$ has even multiplicity. If $2k-2$ is a square, then $k = 2r^2 + 1$ for some integer r.

Proof. This is immediate from Theorem 3.1. In the nonsquare case, the two boundary values are algebraic conjugates, so their multiplicities in the integral characteristic polynomial agree. In the square case, $2k - 2 = s^2$ forces s even. The exact scalar audit is `scripts/verify_equality_boundary.py`. □

Jørgensen's 9-regular order-96 graph of girth five is an exact equality case:

$$\delta^* = 9, \qquad \lambda_{\min}(D) = -9,$$

with multiplicity eight. The construction is due to Jørgensen [14]; the three local graph representations, handwritten graph6 decoder, characteristic polynomials, root intervals, and provenance boundary are checked by `scripts/verify_proof_audit_09_jorgensen96.py`.

9.1. **A prime-field obstruction.** For an odd prime $q \geq 7$ and $1 \leq m \leq q$, define $G(q, m)$ on vertices $P_{i,j}, Q_{k,\ell}$, with $0 \leq i, k < m$ and $j, \ell \in \mathbb{F}_q$, by

$$\begin{aligned} P_{i,j} &\sim P_{i,j\pm1}, \\ Q_{k,\ell} &\sim Q_{k,\ell\pm2}, \\ P_{i,j} &\sim Q_{k,ik+j}. \end{aligned}$$

This is a balanced specialization of known finite-field girth-five constructions [15]. It is $(m+2)$-regular. A coordinate common-neighbour calculation shows that it has no triangle or 4-cycle for $q \geq 7$: for a cross pair, the possible same-side common neighbours require residues in the disjoint sets $\{\pm1\}$ and $\{\pm2\}$.

Theorem 9.2. *If $G(q, m)$ has diameter three, then it is not a strict counterexample to WOW-284.*

Proof. The zero Fourier block has eigenvalues $m + 2$, $2 - m$, and 2 with multiplicity $2m - 2$; the shifted WOW window leaves only $m \in \{4, 5, 6\}$. Let $\omega = e^{2\pi i/q}$. On the nonzero character $t = 1$, the adjacency block has form

$$\begin{pmatrix} aI & M \\ M^* & bI \end{pmatrix}, \qquad a = 2\cos\frac{2\pi}{q}, \quad b = 2\cos\frac{4\pi}{q}, \quad M_{ik} = \omega^{ik}.$$

If σ is a singular value of M, the associated two-dimensional invariant subspace carries $\left(\begin{smallmatrix} a & \sigma \\ \sigma & b \end{smallmatrix}\right)$. Since $\|M\|_F^2 = m^2$, one singular value satisfies $\sigma^2 \geq m$, and the block has a nonprincipal eigenvalue at least

$$\sqrt{m} + \frac{a+b}{2} \geq \sqrt{m} + \cos(\pi/7) - \frac{1}{2}.$$

Here the last inequality uses $2\cos(2\pi/7) + 2\cos(4\pi/7) = 2\cos(\pi/7) - 1$. Moreover $\cos(\pi/7) > \sqrt{3}/2$, and the increasing function $h(m) = \sqrt{2m+2} - \sqrt{m}$ satisfies

$$h(m) \leq h(6) = \sqrt{14} - \sqrt{6} < \frac{27}{20} < \frac{\sqrt{3}+1}{2}.$$

Thus the displayed nonprincipal eigenvalue lies above $-1 + \sqrt{2m+2}$ for $m = 4, 5, 6$. The common-neighbour case split, Fourier reduction, radical comparisons, and exact $q = 7$ controls are independently checked by `scripts/verify_proof_audit_08_prime_field.py`. □

9.2. **Layer-respecting matching deletions.** For $\pi \in S_5$, delete the perfect matching

$$\mathcal{M}_\pi = \left\{ \{P_{i,j}, Q_{\pi(i), i\pi(i)+j}\} : i, j \in \mathbb{F}_5 \right\}$$

from the Hoffman–Singleton graph.

Theorem 9.3. *Each deletion produces a connected simple* 6*-regular graph of order* 50*, girth five, and diameter four. The* 120 *labelled graphs form exactly two isomorphism classes. The* 20 *affine permutations have*

$$\lambda_{\min}(D) = -13, \qquad \Phi = -7,$$

and the 100 *nonaffine permutations have*

$$\lambda_{\min}(D) = -6 - \sqrt{61}, \qquad \Phi = -\sqrt{61}.$$

Thus every member strictly satisfies the conjectured inequality.

Proof. Every $P_{i,j}$ occurs once in $\mathcal{M}_\pi$; for a fixed $Q_{k,\ell}$, the unique incident matching edge is obtained from $i = \pi^{-1}(k)$ and $j = \ell - ik$. Thus $\mathcal{M}_\pi$ is a perfect matching, and its deletion leaves a simple 6-regular graph. Deleting edges cannot create a short cycle, while the same-layer pentagons remain; exact breadth-first search gives connectedness and diameter four.

Explicit type-preserving and type-swapping coordinate automorphisms generate orbits of sizes 20 and 100. The representatives have different adjacency characteristic polynomials, so the orbits are distinct isomorphism classes. Exact distance characteristic polynomials and Sturm separators give the stated least roots. All 120 matchings, 400 coordinate maps, 48,000 matching images, orbit coverage, graph hypotheses, and root certificates are checked by `scripts/verify_proof_audit_07_layer_matchings.py`. □

10. Exact computation and formal verification

The analytic arguments above are primary. Exact computation is used in three roles: to certify explicitly labelled finite graphs, to exhaust precisely specified finite orbit families, and to check symbolic identities whose written derivations are also supplied. No theorem-level sign or eigenvalue ordering uses floating-point arithmetic. The exact symbolic and graph computations use SymPy and NetworkX [16, 17].

The accompanying release archives the labelled graph data, exact rational and polynomial certificates, and exhaustive finite-orbit records used here. These materials make the finite computations independently reproducible without interrupting the mathematical exposition with a file-by-file index.

The explicit 50-vertex counterexample is fully formalized in Lean 4.31 with Mathlib 4.31 [18, 19]. The development checks the coordinate graph, regularity, the exhaustive common-neighbour certificate, girth five, the adjacency-square and distance-matrix identities, and an exact rational diagonalization with multiplicities. Thus the 50-vertex result is verified at graph level, including its least distance eigenvalue and strict WOW-284 gap.

Lean also kernel-checks finite spectral certificates attached to the explicit constructions of orders $38, 39, 40, 42$. For orders $38, 39, 42$, the endpoints certify the stated minimum dual-degree data, positive definiteness of the corresponding shifted finite distance matrix, and hence the strict bound for every nonzero real eigenpair. For order 40, Lean certifies a two-sided invertible exact diagonalization, the diagonal-entry multiplicities, the attained least eigenvalue -5, dual degree six, and gap one. These non-50 endpoints are finite spectral certificates: they do not identify each semantic matrix with `SimpleGraph.dist` and bundle every structural hypothesis into one public theorem.

Separately, Lean formalizes the analytic optimization statement in Theorem 5.3. For every integer $k \geq 4$ and every admissible finitely supported expansion

$$f = \sum_i c_i F_i, \qquad c_0 > 0, \qquad c_i \geq 0 \quad (i \geq 5), \qquad f|_{I_k} \leq 0,$$

it proves

$$B_k c_0 \leq f(k), \qquad B_k = \frac{(k+2)(k^2+3)}{6}.$$

The formal development defines the coefficient family of the displayed quartic f_*, proves that it is admissible and attains equality, and proves that every equality case is its unique positive scalar multiple, both as a polynomial and at coefficient level. The kernel-checked dependency chain includes the nonbacktracking recurrence, the exact primal expansion, positivity and moment identities for the three-point dual certificate, strict slacks in degrees 5 through 9, the Chebyshev tail for every degree at least 10, finite-support weak duality, complementary slackness, and optimizer rigidity.

This LP formalization is deliberately graph-independent. It does not formalize the trace interpretation of the $F_i(A)$, the girth-five vanishing and nonnegativity statements, the spectral trace decomposition, or the passage from the analytic LP inequality to graph-order bounds. Consequently the graph conclusions $n < B_k$, the degree-six reductions $n \le 51$ and $n \le 50$, and the edge-local cycle argument remain conventional analytic proofs, with exact executable audits where cited. The punctured-Moore spectra, the general deletion-stability inequality, the small-puncture normal form, and the Hoffman–Singleton deletion-radius theorem are likewise analytic results supported by exact Python audits; they are not part of the Lean claim.

The release audit rejects `sorry`, `admit`, `native_decide`, `bv_decide`, unsafe declarations, and new axioms. The public endpoint axiom reports contain only `propext`, `Classical.choice`, and `Quot.sound`.

11. Scope, literature boundary, and open questions

Several ingredients belong to established theory and should not be conflated with the project-derived statements.

- Moore graphs, the Hoffman–Singleton graph, and their intact adjacency and distance spectra are classical [3, 20].
- Proposition 2.1 is Backelin's second-degree identity in normalized form [7].
- Proposition 3.3 belongs to the established distance-polynomial framework [4].
- The one-variable nonbacktracking LP framework is due to Nozaki [5]; Theorem 5.3 is the exact two-sided WOW specialization and optimizer theorem derived here.
- The four $(5, 5)$-cages used in Theorem 4.2 come from Meringer's published exhaustive generation [13]; the repository independently verifies the four fixed records but does not replace the enumeration.
- Literature on vertex-deleted adjacency spectra and subconstituents is close in language but does not directly give the recomputed ordinary distance spectra in Section 7 [21, 22].

The main open questions are:

(1) Is there a counterexample of order below 38, and what is the true minimum order?
(2) Does a 6-regular order-50 strict counterexample exist?
(3) Can the one-point and three-vertex constraints of Theorem 6.5 be completed by a multipoint semidefinite or canonical-generation argument?
(4) Do regular diameter-four counterexamples exist for degree at least ten?
(5) Is there an unconditional infinite family of strict counterexamples? Any infinite regular family must have unbounded degree, since Corollary 4.5 and the Moore bound leave finitely many graphs at each fixed degree.
(6) Which deleted sets in a general Moore graph preserve strictness, beyond the metric normal form of Theorem 8.1?

OpenAI ChatGPT-5.6 Sol Pro assisted with adversarial proof checking, proof exploration, and Lean formalization. No AI system is an author. The author assumes full responsibility for the mathematics, attribution, and conclusions. The source, exact certificates, and build instructions are available at `github.com/SamPetkov/wow284` and correspond to release `v2.2.2`.

Appendix A. Characteristic polynomials for the order-39 and order-38 descendants

For every labelled singleton deletion of R,

$$\begin{aligned} P_{39}(x) = {} & x^9(x+5)^{12}(x^2+6x+3) \\ & \cdot (x^3+3x^2-15x-7)^2(x^3+3x^2-15x-3)^2 \\ & \cdot (x^4-78x^3+303x^2-70x-450). \end{aligned}$$

For every labelled edge-endpoint deletion of R,

$$\begin{aligned} P_{38}(x) = x^4(x-2)(x+5)^8(x^2+6x+2)^2 \\ \cdot (x^3+3x^2-15x-3) \\ \cdot (x^4+5x^3-7x^2-23x-6) \\ \cdot (x^5+9x^4+7x^3-77x^2-54x-4) \\ \cdot (x^9-67x^8-404x^7+1772x^6+7205x^5 \\ -18489x^4-17018x^3+20288x^2+16824x+1680). \end{aligned}$$

The fixed graph6, adjacency-list, edge-list, and distance-matrix records are in `data/graphs/`.

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